

UNIT II

Speech Recognition

Content

- Real and Complex Cepstrum
- Application of cepstral analysis to speech signal
- Feature extraction for speech
- MFCC
- LPCC
- Static and dynamic feature for speech recognition
- Distance measures
- Vector quantization models
- HMM
- Robustness issues
- Discrimination in the feature space
- Feature selection
- Gaussian Mixture model

Content

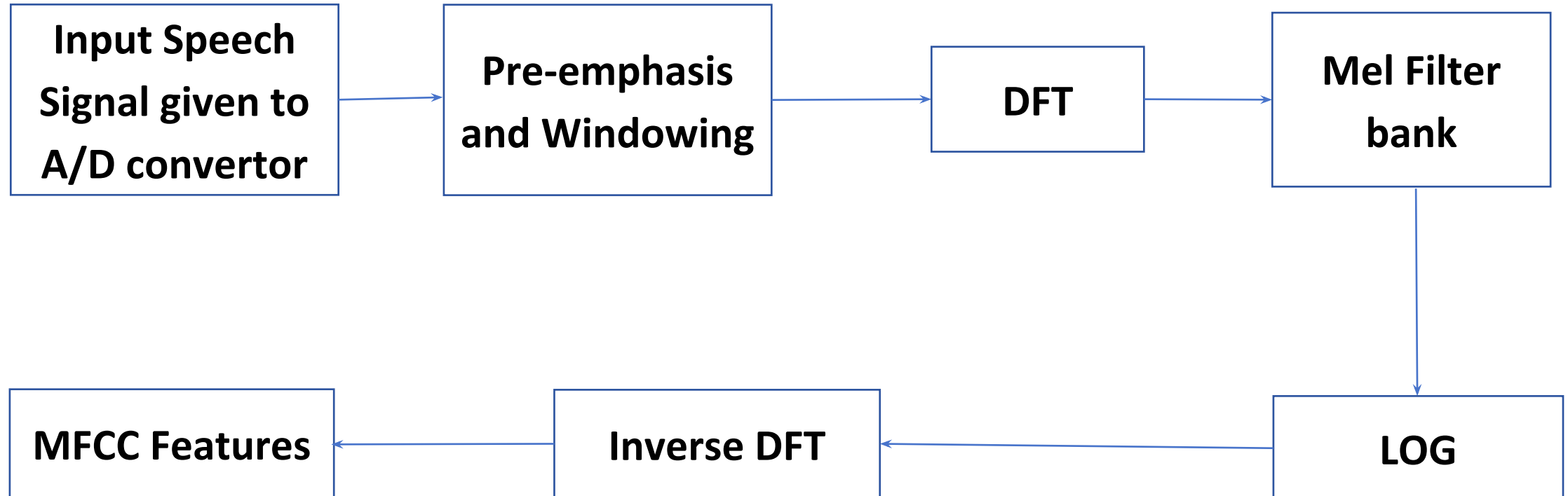
- Mel-Frequency Cepstral Coefficients (MFCCs)
 - Steps to obtain MFCC
 - Applications of MFCC

MFCC

- Mel-Frequency Cepstral Coefficients (MFCCs) is the most widely used feature extraction technique in traditional speech processing.
- Used to represent the power spectrum of an audio signal based on how the human ear actually perceives sound.
- Humans do not perceive pitch linearly.
- Humans are much better at detecting small changes in low frequencies than in high frequencies.
- MFCC provided a compact matrix (typically 12–13 coefficients per time frame) that captures the fingerprint of the sound.

MFCC

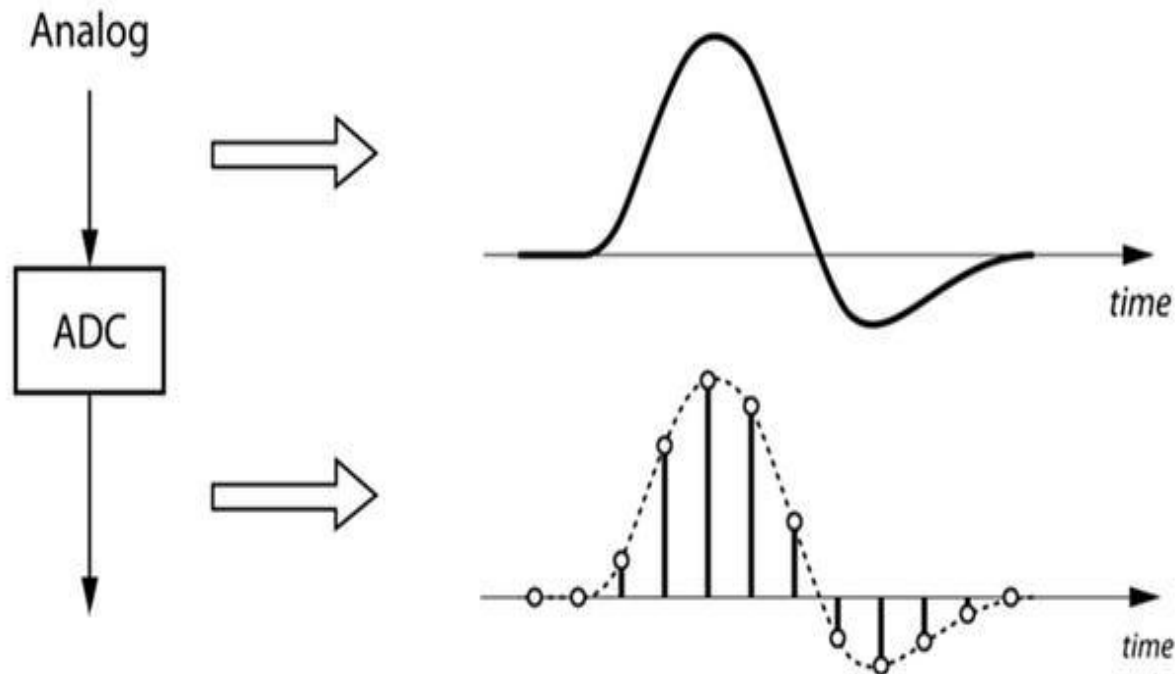
- Steps to extract MFCC:



A/D Conversion

1. A/D Conversion:

- A/D conversion samples the audio clips and digitizes the content, i.e. converting the analog signal into discrete space.
- A sampling frequency of 8 or 16 kHz is often used.



Pre-emphasis

2. Pre-emphasis:

- Pre-emphasis boosts the amount of energy in the high frequencies.
- For voiced segments like vowels, there is more energy at the lower frequencies than the higher frequencies.
- This is called spectral tilt which is related to the glottal source (how vocal folds produce sound).
- Boosting the high-frequency energy makes information in higher formants more available to the acoustic model.
- Pre-emphasis is used to make the system less susceptible to noise introduced in the process later.

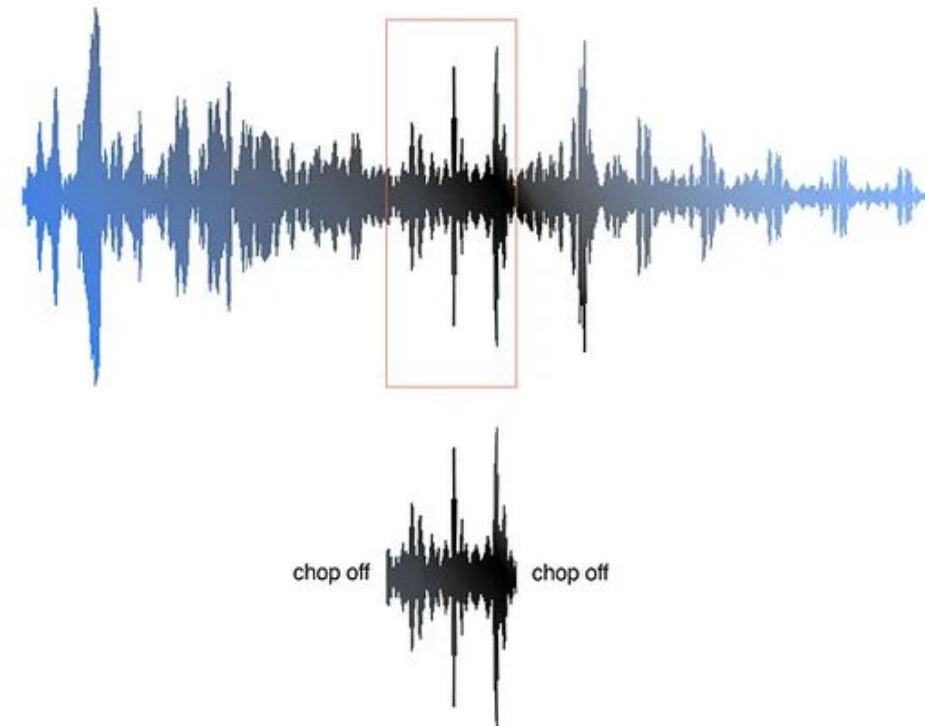
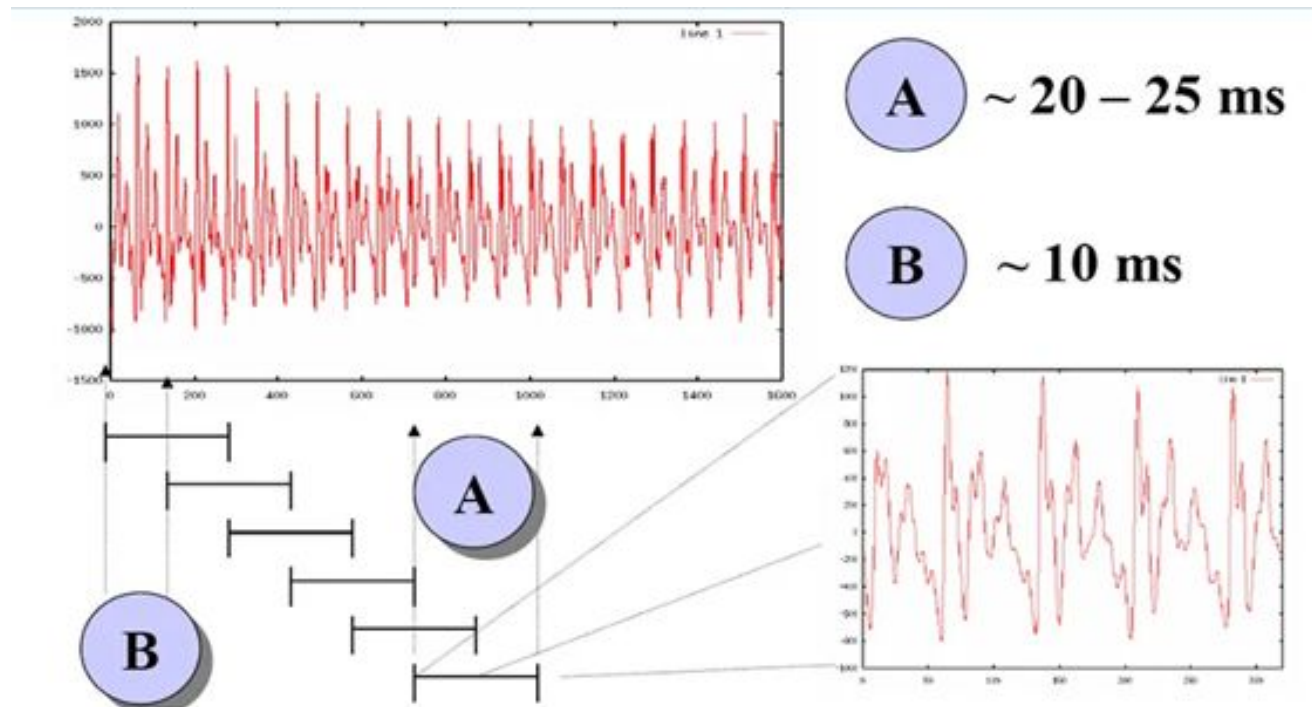
Windowing

3. Windowing :

- Windowing involves the slicing of the audio waveform into sliding frames.
- It cannot be just chop it off at the edge of the frame.
- The suddenly fallen in amplitude will create a lot of noise that shows up in the high-frequency.
- To slice the audio, the amplitude should gradually drop off near the edge of a frame.

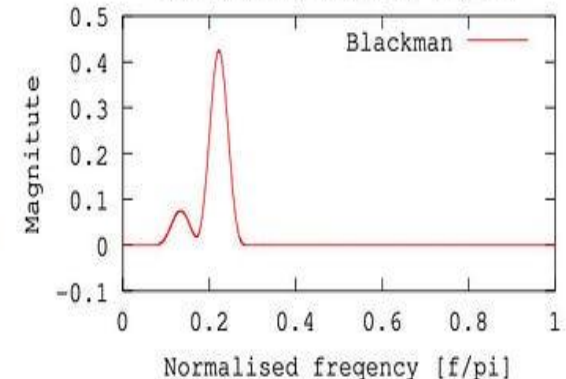
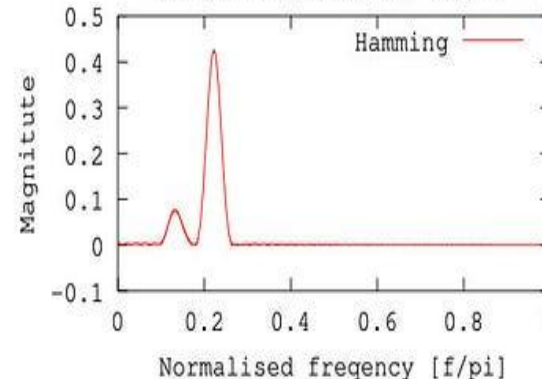
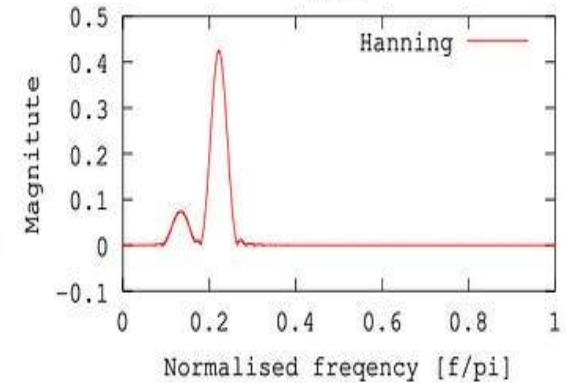
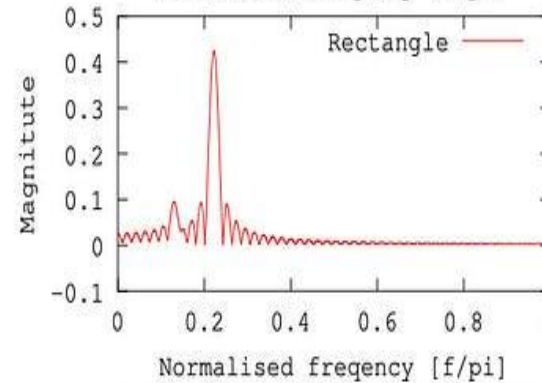
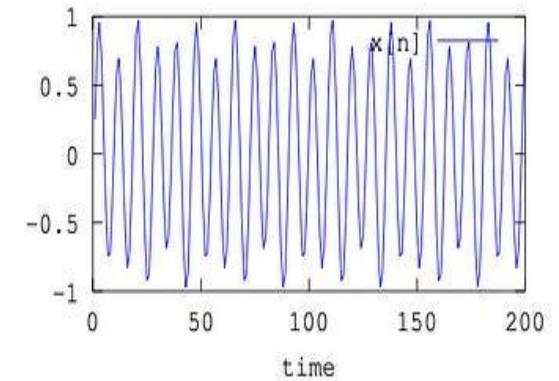
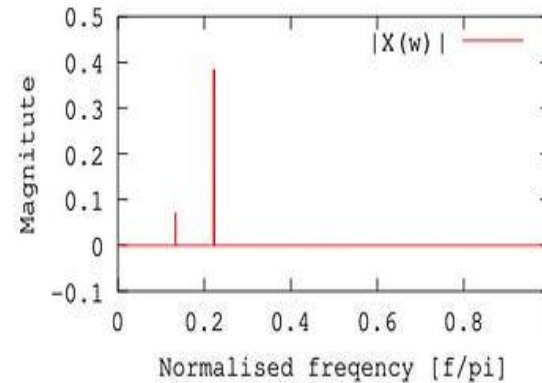
Windowing

- It can be written as $x[n] = w[n] s[n]$
- w is the window applied to the original audio clip in the time domain.
- w may be the Hamming window and the Hanning window.



Windowing

- The chopped frame with Hamming and Hanning maintains the original frequency information better with less noise compared to a rectangle window.



Discrete Fourier Transform

4. Discrete Fourier Transform (DFT)

- Apply DFT to extract information in the frequency domain.
- DFT of signal $x[n]$ is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

Mel filterbank

5. Mel filterbank

- The equipment measurements are not the same as our hearing perception.
- For humans, the perceived loudness changes according to frequency.
- Also, perceived frequency resolution decreases as frequency increases i.e. humans are less sensitive to higher frequencies.
- The Mel scale maps the measured frequency to that we perceived in the context of frequency resolution.
- All these mappings are non-linear.
- Apply triangular band-pass filters to convert the frequency information to mimic what a human perceived.

Mel filterbank

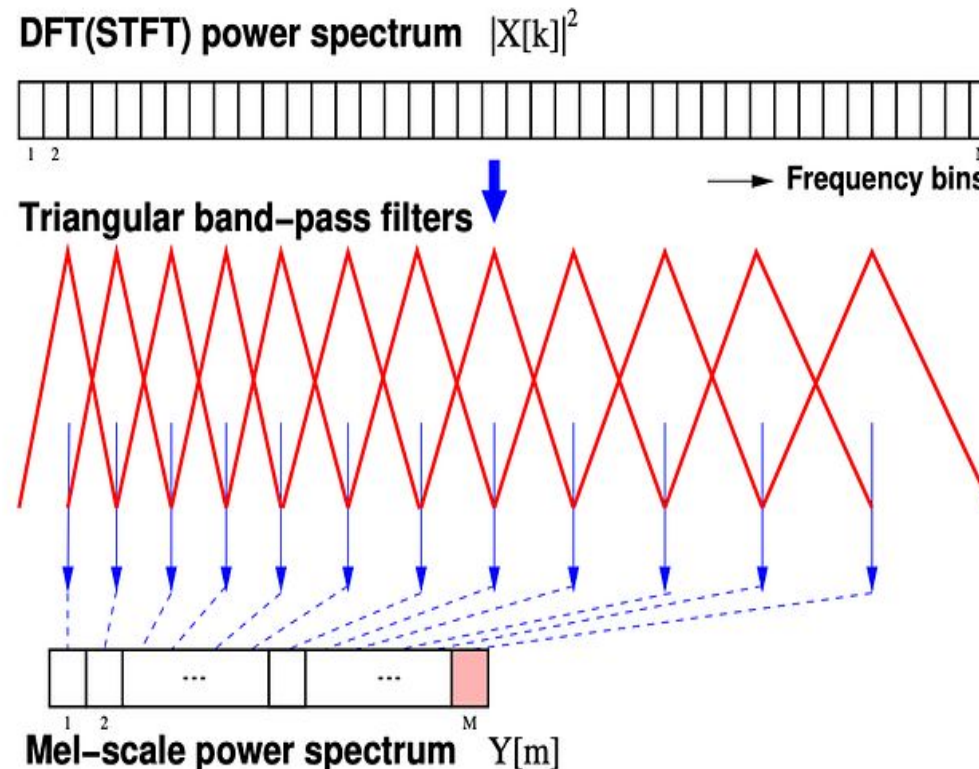
- Square the output of the DFT that reflects the power of the speech at each frequency ($|x[k]|^2$) and it is the DFT power spectrum.
- Triangular Mel-scale filter banks transform it to Mel-scale power spectrum.
- The output for each Mel-scale power spectrum slot represents the energy from a number of frequency bands that it covers. This mapping is called the Mel Binning.
- The precise equations for slot m will be:

$$Y_t[m] = \sum_{k=1}^N W_m[k] |X_t[k]|^2$$

where k : DFT bin number $(1, \dots, N)$
 m : mel-filter bank number $(1, \dots, M)$

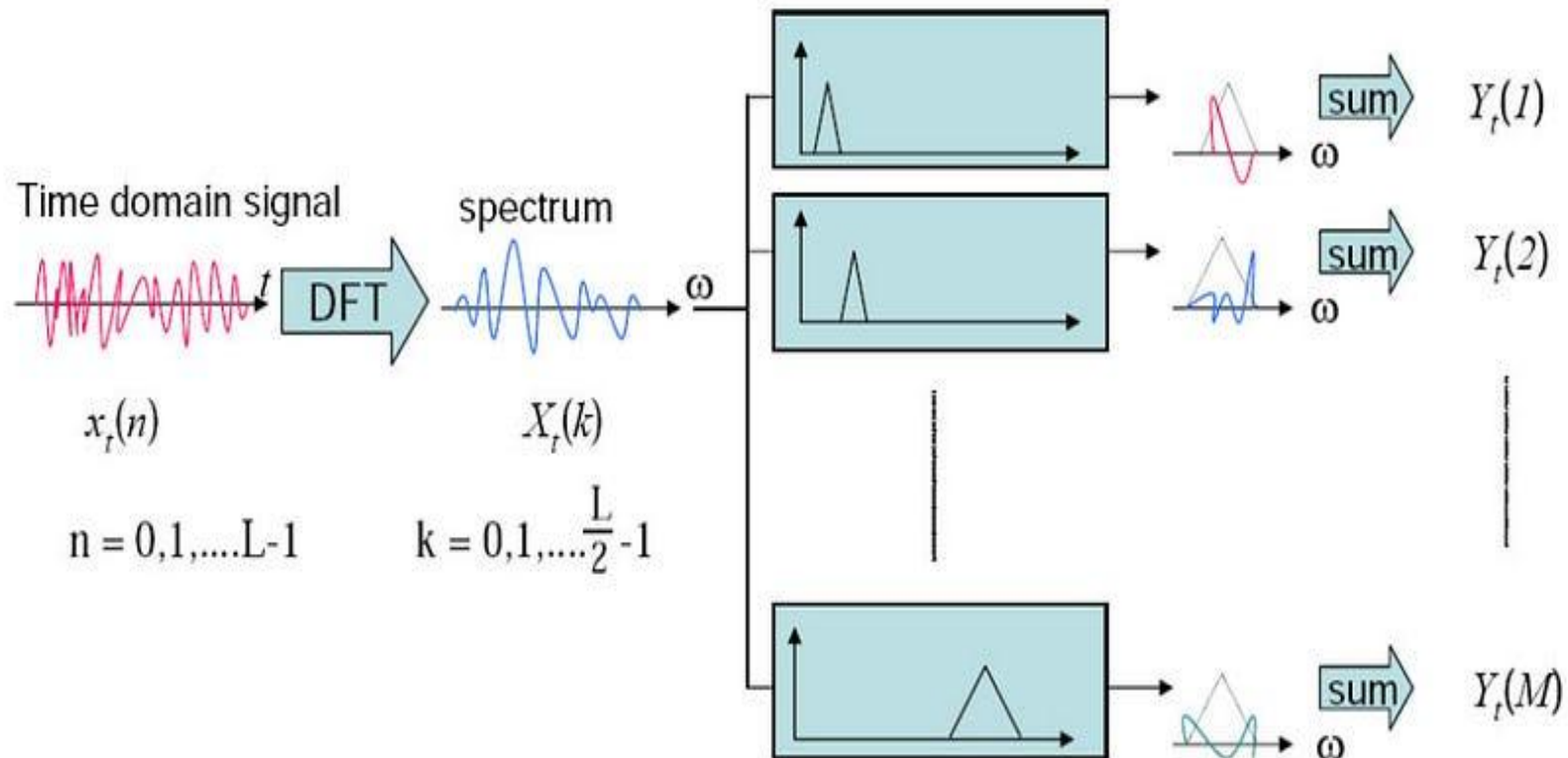
Mel filterbank

- The Triangular bandpass is wider at the higher frequencies to reflect human hearing is less sensitivity in high frequency.
- All these efforts try to mimic how the basilar membrane in ear senses the vibration of sounds.



Mel filterbank

- The basilar membrane has about 15,000 hairs inside the cochlear at birth.
- The frequency response of the hairs is approximated by triangles in Mel filterbank.
- It imitate how ears perceive sound through those hairs.



Log

6. Log:

- Mel filterbank outputs a power spectrum.
- Humans are less sensitive to small energy change at high energy than small changes at a low energy level.
- It is logarithmic.
- Take the log out of the output of the Mel filterbank.
- It reduces the acoustic variants that are not significant for speech recognition.

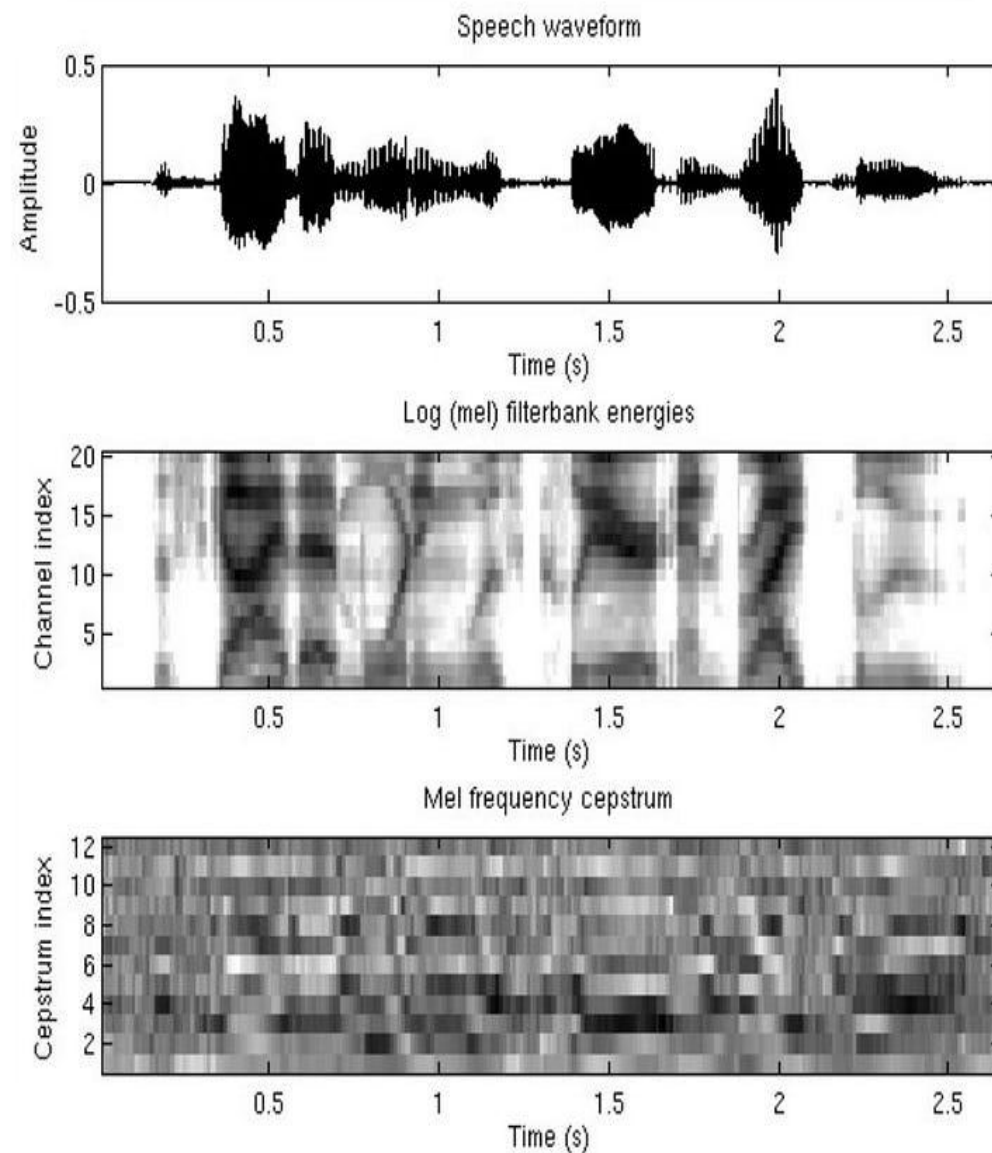
IDFT

7. IDFT:

- Inverse Fourier Transformation is taken to separate the pitch information from the formants.
- MFCC just takes the first 12 cepstral values and discard others.

IDFT

- Visualization of the 12 Cepstrum coefficients.



Applications of MFCC

- Automated Speech Recognition (ASR): Converts spoken language into text (used in virtual assistants like Siri, Alexa).
- Phoneme Classification: Identifies individual structural sounds (like "sh" or "ah") to map out spoken words.
- Keyword Spotting: Powers voice-activation triggers like "Hey Siri" or "OK Google" using low computational footprint pipelines.
- Speaker Identification: Recognizes who is speaking out of a known database by capturing the unique physical structure of their vocal tract.
- Speaker Verification: Acts as a security layer to verify a user's voiceprint passkey for banking or secure device access.